A conditional noncommutative Poisson–Carleson decomposition via stopped random walks

October 21, 2026

This note continues Remark `cd Mei`(ii) of `nc-Davis_v3.tex`. Its only nonstandard input is the following endpoint Davis estimate. The argument below does not prove that input. In particular, this is a consequence conditional on the stated version of the manuscript's main theorem, rather than an independent proof of it.

Assumption 1 (Davis estimate for regular filtrations)**.** *There is a constant D_4, independent of the finite probability von Neumann algebra and the length of the filtration, such that*

$$\|x\|_{H_1} \le D_4 \, \|(\mathcal{E}_n x)_n\|_{L_1(\mathcal{A};\ell_\infty)}$$

for every L_2-martingale associated with a 4-regular filtration. The initial algebra is allowed to be nontrivial. We index its conditional expectation by $\mathcal{E}_0$; this is an actual conditional expectation, not the zero map. The Hardy norm includes the initial martingale term $\mathcal{E}_0 x$.

Let $(\mathcal{N},\varphi)$ be a finite von Neumann algebra with normalized faithful normal trace. Put

$$\|f\|_{\mathrm{BMO}_{cr}} = \max\left\{ \sup_I \left\| \fint_I (f-f_I)^*(f-f_I)\,dx \right\|_{\mathcal{N}}^{1/2}, \sup_I \left\| \fint_I (f-f_I)(f-f_I)^*\,dx \right\|_{\mathcal{N}}^{1/2} \right\}.$$

We use Mei's operator-valued BMO space, with bounded operator-valued interval averages, modulo constants. For a positive operator-valued Borel measure ν on $\mathbb{R}^2_+$, weak-star countably additive and finite on bounded rectangles, set

$$\|\nu\|_{\mathrm{Carl}} = \sup_I |I|^{-1} \, \|\nu(I\times(0,|I|])\|_{\mathcal{N}}\,.$$

An order bound $\nu(T(I)) \le A|I|1_{\mathcal{N}}$ is equivalent to this bound. For a difference $\mu = \mu_+ - \mu_-$ of positive measures, its Poisson balayage is

$$S_P\mu(x) = \int_{\mathbb{R}^2_+} P_y(x-t)\,d\mu(t,y), \qquad P_y(s) = \frac{1}{\pi}\frac{y}{s^2+y^2}.$$

Theorem 2. *Assume Assumption 1. If $f = f^* \in \mathrm{BMO}_{cr}(\mathbb{R},\mathcal{N})$ has compact support, there are a selfadjoint $g \in L_\infty(\mathbb{R})\bar{\otimes}\mathcal{N}$ and finite positive $\mathcal{N}$-valued measures μ_+,μ_-, supported in a bounded rectangle of the upper half-plane, such that*

$$\boxed{f = g + S_P(\mu_+ - \mu_-)}$$

and

$$\boxed{\|g\|_\infty + \|\mu_+ + \mu_-\|_{\mathrm{Carl}} \le C(D_4)\,\|f\|_{\mathrm{BMO}_{cr}}\,.}$$

The equality holds in $L_1(\mathbb{R}\bar{\otimes}\mathcal{N})$, and in particular locally in L_1 and distributionally. The constant is independent of the matrix dimension of $\mathcal{N}$. For a general complex f, four positive measures suffice.

1. A signed martingale representation

For the filtration indexed by $n \geq 0$, use the inhomogeneous column martingale BMO norm

$$\max\left\{\|\mathcal{E}_0(F^*F)\|_\infty^{1/2},\ \sup_{n\geq 1}\left\|\mathcal{E}_n|F-\mathcal{E}_{n-1}F|^2\right\|_\infty^{1/2}\right\},$$

and take the maximum with the row norm. This is the manuscript's convention after shifting the first index from 1 to 0.

Lemma 3. *Under Assumption 1, if $F = F^* \in \mathrm{BMO}(\mathcal{A})$, for a finite probability algebra $\mathcal{A}$ with a 4-regular filtration, then*

$$F = G + \sum_{n\geq 0}\mathcal{E}_n(v_n^+ - v_n^-), \qquad G = G^*, \quad v_n^\pm \geq 0,$$

where

$$\|G\|_\infty + \left\|\sum_n (v_n^+ + v_n^-)\right\|_\infty \leq C(D_4)\,\|F\|_{\mathrm{BMO}}\,.$$

Both positive series after conditional expectation converge in L_1.

Proof. For $x \in L_2(\mathcal{A})$, define

$$Jx = (x, \mathcal{E}_0x - x, \mathcal{E}_1x - x, \ldots).$$

The L_2-Doob inequality implies that $Jx \in L_1(\mathcal{A}; c_0)$: indeed, for $x_m = \mathcal{E}_m x$, the maximal L_1-norm of the tail $(\mathcal{E}_n x - x)_{n\geq m}$ is at most $C\,\|x - x_m\|_2$, since the trace is finite and normalized. Also

$$\|(\mathcal{E}_n x)_n\|_{L_1(\ell_\infty)} \leq 2\,\|Jx\|_{L_1(\ell_\infty)}\,.$$

The H_1–BMO duality and Assumption 1 therefore show that $Jx \mapsto \tau(Fx)$ is bounded by $CD_4\,\|F\|_{\mathrm{BMO}}$. Hahn–Banach and

$$L_1(\mathcal{A}; c_0)^* = L_\infty(\mathcal{A}; \ell_1)$$

give a sequence $(w, u_0, u_1, \ldots)$ of that norm with

$$\tau(Fx) = \tau(w^*x) + \sum_n \tau\big(u_n^*(\mathcal{E}_n x - x)\big).$$

The c_0 duality is the endpoint assertion in Junge–Xu, *Noncommutative maximal ergodic theorems*, Remark 2.2(ii). After replacing the representing sequence by its selfadjoint part, write

$$F = G + \sum_n \mathcal{E}_n u_n, \qquad G = w - \sum_n u_n.$$

The latter sum converges weak-star with bounded norm.

For completeness, positivity with a bounded sum follows from the factorization defining $L_\infty(\ell_1)$, not from termwise spectral positive parts. Choose balanced factors

$$u_n = \sum_k a_{kn}^* b_{kn}, \qquad \left\|\sum_{k,n} a_{kn}^* a_{kn}\right\|_\infty,\ \left\|\sum_{k,n} b_{kn}^* b_{kn}\right\|_\infty \leq CD_4\,\|F\|_{\mathrm{BMO}}\,.$$

Polarization gives

$$u_n = \sum_{r=0}^{3} i^{-r} v_n^{(r)}, \qquad v_n^{(r)} = \frac{1}{4} \sum_k (a_{kn} + i^r b_{kn})^* (a_{kn} + i^r b_{kn}) \geq 0.$$

Each $\sum_n v_n^{(r)}$ is bounded by the same constant up to an absolute factor. Taking real parts gives $u_n = v_n^{(0)} - v_n^{(2)}$. These positive sequences have bounded sums; their conditional expectations have summable traces, proving L_1 convergence. □

2. Scalar discrete harmonic kernels

Take $h = 1/(2m)$. Let a simple symmetric nearest-neighbor random walk on $h\mathbb{Z}^2$ move in the four coordinate directions with probability $1/4$. Its horizontal starting point may be any real number. Let Q_{jh}^h be the horizontal exit-distribution operator when the walk starts at height jh and is stopped on height 0. Thus

$$Q_{jh}^h a(x) = \sum_{k \in \mathbb{Z}} p_j(k) a(x + kh), \qquad Q_0^h = I.$$

The kernels are symmetric probability distributions and $Q_{jh}^h Q_{lh}^h = Q_{(j+l)h}^h$.

Lemma 4. *There is an absolute C such that, for $j \geq 1$,*

$$p_j(k) \leq C \frac{j}{j^2 + k^2}.$$

For every Schwartz function ψ,

$$\sup_{\substack{0 \leq y \leq 1 \\ y \in h\mathbb{Z}}} \sup_x |Q_y^h \psi(x) - P_y \psi(x)| \longrightarrow 0.$$

Moreover, if I has length L, $t \in I$, and $0 \leq y \leq L$, then an unrestricted walk started at (t, y) exits the lower boundary inside DI with probability at least $1/2$, for an absolute dilation D.

Proof. The horizontal Fourier multiplier is

$$\sum_k p_j(k) e^{ik\theta} = \exp[-ja(\theta)], \qquad a(\theta) = \operatorname{arcosh}(2 - \cos\theta).$$

This follows from the discrete harmonic equation $r + r^{-1} = 4 - 2\cos\theta$, choosing the root $0 < r \leq 1$. On $[-\pi, \pi]$, $a(\theta) \simeq |\theta|$; its first derivative is bounded on each side of zero and its second derivative is bounded there. The integral formula gives $p_j(k) \leq C/j$. Integrating by parts twice on the two sides of zero gives $p_j(k) \leq Cj/k^2$: the jump of the first derivative contributes $O(j)$, and the L_1-norm of the second derivative of e^{-ja} is $O(j)$. Together these imply the displayed kernel bound.

Extend a periodically with period 2π. Writing $a_h(\xi) = h^{-1} a(h\xi)$, the multiplier of Q_y^h is $e^{-y a_h(\xi)}$, and $a_h(\xi) \to |\xi|$ for fixed ξ. Fourier inversion and dominated convergence yield uniform convergence for $0 \leq y \leq 1$. Finally $\sum_{|k| > Aj} p_j(k) \leq C/A$; choose A large and then D large enough. The assertion for $y = 0$ is immediate. □

3. Uniform BMO control of the stopped lift

Translations and dilations reduce the theorem to $\operatorname{supp} f \subset [-1/8, 1/8]$. Write $K = \|f\|_{\mathrm{BMO}_{cr}}$; the case $K = 0$ is trivial. Averages over $[-1/2, 1/2]$, using that f vanishes on a fixed fraction of this interval, give the order bounds

$$\int_{\mathbb{R}} f^* f \, dx \le CK^2 1_{\mathcal{N}}, \qquad \int_{\mathbb{R}} f f^* \, dx \le CK^2 1_{\mathcal{N}}. \tag{11}$$

Indeed, if Q is this interval and $|\operatorname{supp} f| \le |Q|/2$, operator Cauchy–Schwarz gives $|f_Q|^2 \le \frac{1}{2} \fint_Q f^* f$; combine this with $\fint_Q |f - f_Q|^2 \le K^2 1$. The row estimate is identical.

Set $f_\epsilon = P_\epsilon f$. It is bounded and operator-norm Lipschitz, its BMO norm is at most K, and it satisfies the same integrated square bounds (11). We use the elementary Poisson-variance estimate

$$\sup_{x,y} \left\| P_y\big(|f_\epsilon - P_y f_\epsilon(x)|^2\big)(x) \right\|_{\mathcal{N}} \le CK^2. \tag{12}$$

This follows by decomposing the Poisson integral into concentric dyadic intervals: the square oscillation on the r-th interval relative to the smallest interval average is at most $C(r+1)^2 K^2$, and the Poisson weights decrease as 2^{-r}. The mean $P_y f_\epsilon(x)$ minimizes this operator-valued variance. The same estimate holds for adjoints.

Choose h small enough that

$$h \operatorname{Lip}(f_\epsilon) \le K.$$

Comparing $p_j(k)$ with the Poisson mass of the interval of length h centered at kh, and using the Lipschitz bound, yields

$$Q_y^h |f_\epsilon - c|^2(x) \le C P_y |f_\epsilon - c|^2(x) + Ch^2 \operatorname{Lip}(f_\epsilon)^2 1_{\mathcal{N}}. \tag{13}$$

In particular (12)–(13) imply

$$\left\| Q_y^h |f_\epsilon|^2(x) - |Q_y^h f_\epsilon(x)|^2 \right\|_{\mathcal{N}} \le CK^2. \tag{14}$$

Start the walk at $(X_0, 1/2)$, with X_0 uniform on $[-R, R]$. Let τ be its first exit from $0 < y < 1$, and thereafter keep the position $Z_n = (X_n, Y_n)$ fixed. The filtration reveals X_0 initially and then all four-way coins, including after τ. It is 4-regular. For a boundary function a, define

$$T_h a = \begin{cases} a(X_\tau), & Y_\tau = 0, \\ Q_1^h a(X_\tau), & Y_\tau = 1. \end{cases}$$

Discrete harmonicity and optional stopping give exactly

$$\mathbb{E}_n(T_h a) = Q_{Y_n}^h a(X_n). \tag{15}$$

Put $F = T_h f_\epsilon$. Its conditional future variance is bounded by (14): it is the stopped conditional expectation of the unrestricted lower-exit terminal value $f_\epsilon(\xi)$. Thus its little BMO square norm is $O(K^2)$. Regularity controls the big BMO terms by four times these terms. The initial term is controlled by

$$\mathbb{E}_0 |F|^2 \le Q_{1/2}^h |f_\epsilon|^2(X_0) \le CK^2 1_{\mathcal{N}},$$

using (11), Jensen for P_ϵ, and (13). Consequently

$$\|F\|_{\mathrm{BMO}} \le CK \tag{16}$$

uniformly in R, ϵ, h subject to the preceding small-mesh condition.

4. Transfer of the representation and its Carleson bound

Apply Lemma 3 to F:

$$F = G + \sum_n \mathbb{E}_n(v_n^+ - v_n^-), \qquad \|G\|_\infty \le A, \quad 0 \le \sum_n (v_n^+ + v_n^-) \le A1_{\mathcal{N}}, \quad A = C(D_4)K.$$

Define positive measures on the closed strip by

$$\mu_{R,h}^{\pm}(B) = 2R\,\mathbb{E}\sum_n v_n^{\pm}\,\mathbf{1}_{\{Z_n \in B\}}, \qquad B \subset \mathbb{R} \times [0,1]. \tag{17}$$

The variables $v_n^{\pm}$ need not be adapted; only their bounded positive sum is used.

For every interval I,

$$(\mu_{R,h}^+ + \mu_{R,h}^-)(I \times [0,|I|]) \le CA|I|1_{\mathcal{N}}. \tag{18}$$

To see this, let σ be the first visit of the stopped path to that box. If $\sigma < \infty$, it is at or before strip exit. From Z_σ, continue an unrestricted walk to the lower boundary. By Lemma 4, its exit lies in DI with conditional probability at least $1/2$. Hence

$$2R\,\mathbb{P}(\sigma < \infty) \le 2(2R)\mathbb{P}(\xi \in DI) \le 2D|I|.$$

The last inequality holds because convolution of the initial density $\mathbf{1}_{[-R,R]}$ with any probability kernel is at most 1. Equation (18) now follows from the pointwise positive-sum bound.

Test the martingale representation against $T_h\psi$, and use (15):

$$2R\,\mathbb{E}\big[(T_h f_\epsilon)^* T_h\psi\big] = \int_{\mathbb{R}} g_{R,h}(x)^*\psi(x)\,dx + \int Q_y^h\psi(t)\,d(\mu_{R,h}^+ - \mu_{R,h}^-)(t,y). \tag{19}$$

This is an operator-valued weak identity, or equivalently can be tested against every normal functional on $\mathcal{N}$. The trace notation in deriving it is suppressed in (19). The function $g_{R,h}$ is the adjoint image of G. It satisfies $\|g_{R,h}\|_\infty \le A$: under Lebesgue-distributed initial horizontal positions the two strip exit marginals are each $\frac{1}{2}dx$; truncating initial positions only decreases them. The top marginal is then convolved with the probability kernel Q_1^h.

Let $R \to \infty$. The two exit marginals increase to $\frac{1}{2}dx$. Symmetry and the vertical semigroup property give

$$\lim_{R\to\infty} 2R\,\mathbb{E}\big[(T_h f_\epsilon)^* T_h\psi\big] = \frac{1}{2}\int_{\mathbb{R}} (f_\epsilon + Q_2^h f_\epsilon)^*\psi\,dx. \tag{20}$$

This extra $Q_2^h f_\epsilon$ term is essential.

The bounds (18) give local weak-star compactness of the positive operator-valued measures; bounded $g_{R,h}$ have weak-star compactness in L_∞. Subnets may be used if the predual is nonseparable. Pass first in R, and then take $\epsilon \downarrow 0$, $h \downarrow 0$ with $h\,\mathrm{Lip}(f_\epsilon) \le K$. The kernel convergence in Lemma 4, together with $f_\epsilon \to f$ and $Q_2^h f_\epsilon \to P_2 f$ in L_2, yields

$$\frac{1}{2}(f + P_2 f) = g_0 + S_P(\lambda_+ - \lambda_-). \tag{21}$$

At height zero the Poisson operator in this formula is understood as the identity.

Here the limiting positive measures are supported in the closed strip and still satisfy (18). For clarity, passage from local weak-star convergence to (21) has no missing tail condition: for a compactly supported smooth test function ψ,

$$|Q_y^h\psi(t)| + |P_y\psi(t)| \le C_\psi(1+t^2)^{-1} \quad (|t| \text{ large},\ 0 \le y \le 1),$$

uniformly in small h. The measure of each horizontal annulus of radius $r \geq 1$, including all heights, is $O(Ar)$. Thus its contribution to the pairing is $O(A/r)$, summable over dyadic annuli. On compact sets kernel convergence is uniform. Open-box approximation passes (18) to the limit.

The height-zero parts obey $\lambda_{\pm}(B \times \{0\}) \leq CA|B|1_{\mathcal{N}}$. They have weak-star measurable densities in $L_\infty(\mathbb{R}) \bar{\otimes} \mathcal{N}$ of norm $O(A)$, and are absorbed into g_0. Also $\|P_2 f\|_\infty \leq CK$, by (11). Multiplying (21) by two and absorbing these terms proves

$$f = g_1 + S_P(\mu_+ - \mu_-), \qquad \|g_1\|_\infty + \|\mu_+ + \mu_-\|_{\mathrm{Carl}} \leq C(D_4)K, \tag{22}$$

with measures supported in $0 < y \leq 1$. At this stage the measures need not have finite total mass.

5. Spatial localization and finite mass

Choose $0 \leq \chi \leq 1$, smooth and compactly supported, equal to 1 on the support of f, with absolute bounds on χ, χ' in the normalized coordinates. If $\mu = \mu_+ - \mu_-$ is as in (22), put

$$E_\chi(x) = S_P(\chi\mu)(x) - \chi(x) S_P\mu(x) = \int P_y(x-t)(\chi(t) - \chi(x))\, d\mu(t,y).$$

Then

$$\|E_\chi\|_\infty \leq C\, \|\mu_+ + \mu_-\|_{\mathrm{Carl}}\,. \tag{23}$$

Indeed, write $r = \max(|t-x|, y)$. On $r \leq 1$, the absolute scalar kernel in this integral is $O(1)$, and the positive dominating measure on $r \leq 2^{-j}$ is $O(A2^{-j})$. On $r \geq 1$, since $y \leq 1$, the kernel is $O(r^{-2})$ and the measure on $r \leq 2^{j+1}$ is $O(A2^j)$. Both geometric series converge. The estimates are operator-order estimates with the positive measure $\mu_+ + \mu_-$.

Since $\chi f = f$, (22)–(23) imply

$$f = \underbrace{\chi g_1 - E_\chi}_{g} + S_P\big(\chi\mu_+ - \chi\mu_-\big).$$

The new measures have compact horizontal support and heights at most 1, hence finite total mass by the Carleson bound. For finite $\mathcal{N}$, their positive balayages are globally in L_1; the final identity therefore holds in L_1. This proves Theorem 2. For complex f, apply the selfadjoint result to its real and imaginary parts. Undoing the spatial normalization completes the general compact support case.

6. General BMO and a renormalized Poisson balayage

The compact support restriction can be removed if one normalizes the far field. Put $I_0 = [-1/2, 1/2]$, $r(t,y) = \max(|t|, y)$, and define

$$\widetilde{S}_P\mu(x) = \int_{\mathbb{R}^2_+} \left[P_y(x-t) - \mathbf{1}_{\{r(t,y)\geq 1\}} P_y(-t)\right]\, d\mu(t,y). \tag{24}$$

Near finite points this is interpreted in local L_1; at infinity the difference is integrated in operator norm on each compact set of x's. Thus (24) does not subtract a potentially divergent global constant as a separate integral.

Corollary 5. *Under Assumption 1, for every selfadjoint $f \in \mathrm{BMO}_{cr}(\mathbb{R}, \mathcal{N})$, there are selfadjoint $g \in L_\infty$ and positive operator-valued measures μ_+, μ_-, finite on bounded rectangles, such that*

$$f - f_{I_0} = g + \widetilde{S}_P(\mu_+ - \mu_-), \qquad \|g\|_\infty + \|\mu_+ + \mu_-\|_{\mathrm{Carl}} \le C(D_4)\, \|f\|_{\mathrm{BMO}_{cr}}. \tag{25}$$

The equality holds locally in L_1, and modulo constants it is independent of the reference normalization in (24).

Proof. Write $K = \|f\|_{\mathrm{BMO}_{cr}}$. Choose a real smooth cutoff η_L, equal to 1 on $[-L, L]$, supported in $[-2L, 2L]$, taking values in $[0, 1]$, and satisfying $\|\eta_L'\|_\infty \le C/L$. For $J_L = [-4L, 4L]$, put

$$f_L = \eta_L(f - f_{J_L}), \qquad L \ge 1.$$

Then $\|f_L\|_{\mathrm{BMO}_{cr}} \le CK$, uniformly in L. Here are the details needed for the operator-valued cutoff estimate. For an interval I of length at least L,

$$⨍_I f_L^* f_L \le |I|^{-1} \int_{J_L} |f - f_{J_L}|^2 \le 8K^2 1_{\mathcal{N}}.$$

If $\ell = |I| < L$ and I meets the support of η_L, then $I \subset J_L$. Nested interval averaging gives

$$\|f_I - f_{J_L}\|_{\mathcal{N}} \le CK\big(1 + \log(L/\ell)\big).$$

With $c = \eta_L(C_I)(f_I - f_{J_L})$, split

$$f_L - c = \eta_L(f - f_I) + (\eta_L - \eta_L(C_I))(f_I - f_{J_L}).$$

The inequality $|a + b|^2 \le 2|a|^2 + 2|b|^2$ bounds its average square by

$$CK^2 \left[1 + (\ell/L)^2\big(1 + \log(L/\ell)\big)^2\right] 1_{\mathcal{N}} \le CK^2 1_{\mathcal{N}}.$$

Variance about the actual interval mean is smaller. Intervals disjoint from the cutoff support are trivial. The row argument is identical.

Apply Theorem 2 to f_L. Obtain

$$f_L = g_L + S_P(\mu_{L,+} - \mu_{L,-}), \qquad \|g_L\|_\infty + \|\mu_{L,+} + \mu_{L,-}\|_{\mathrm{Carl}} \le C(D_4)K. \tag{26}$$

Extract locally weak-star convergent subnets of the positive measures on the closed upper half-plane, and a weak-star convergent subnet of g_L. For every compactly supported smooth ψ with $\int \psi = 0$, and $r = \max(|t|, y)$ sufficiently large,

$$|P_y\psi(t)| = \left| \int \big[P_y(t - x) - P_y(t)\big]\psi(x)\, dx \right| \le C_\psi r^{-2}. \tag{27}$$

The Carleson bound gives mass $O(Kr)$ on each dyadic annulus of radius r. Therefore the tail in pairing (26) with ψ is $O(K/R)$ beyond radius R, uniformly in L. On compact sets, $P_y\psi(t)$ extends continuously to $y = 0$, with value $\psi(t)$. Since $f_L = f - f_{J_L}$ on the support of ψ for large L, the left side pairs exactly as f.

The boundary parts of the limiting measures have bounded densities of norm $O(K)$, by the same argument as after (21); absorb them into the limiting bounded function. Denote the remaining

upper half-plane measures by μ_+, μ_-, and the bounded function by g_0. They have the required uniform bounds. We have proved

$$\langle f - g_0, \psi \rangle = \int P_y \psi(t)\, d(\mu_+ - \mu_-)(t, y) \quad \text{whenever} \quad \int \psi = 0. \tag{28}$$

Formula (24) defines a local L_1 function. Indeed, on a bounded region in (t, y), positivity and finite trace give local L_1 integrability of the ordinary Poisson integral. The subtracted kernel there is bounded on $r \geq 1$. At large r, the mean value theorem bounds the difference kernel by $C_B r^{-2}$, uniformly for x in a compact set B; the annular Carleson estimate sums this in operator norm. Consequently (28) implies

$$f = g_0 + \widetilde{S}_P(\mu_+ - \mu_-) + c$$

for a constant operator c. This constant is bounded after normalizing $f_{I_0} = 0$. To check its bound, the average over I_0 of the renormalized balayage has norm $O(K)$: on $r \leq 2$, use $\fint_{I_0} P_y(x-t)\, dx \leq |I_0|^{-1}$, the boundedness of the subtracted kernel, and the $O(K)$ mass of this region; on $r \geq 2$, use the $O(r^{-2})$ difference bound and annular summation. Thus $\|c\|_{\mathcal{N}} \leq CK$. Absorb c into g_0, proving (25). The same tail estimates justify the stated interpretation of (24). □

Scope and references

The proof uses signed measures even for selfadjoint or positive f. It does not assert the stronger positivity of both g and μ. It also does not use a dimension-independent implication from an arbitrary positive operator-valued Carleson measure to column/row BMO; that would be a different assertion.

- T. Mei, *Operator valued Hardy spaces*, Mem. Amer. Math. Soc. 188 (2007), no. 881. arXiv:math/0311160.
- M. Junge and Q. Xu, *Noncommutative maximal ergodic theorems*, J. Amer. Math. Soc. 20 (2007), 385–439, especially Remark 2.2(ii). arXiv:math/0505308.
- J. B. Garnett and P. W. Jones, *BMO from dyadic BMO*, Pacific J. Math. 99 (1982), 351–371. The random-walk argument here replaces the kernel-replacement step of the scalar proof.